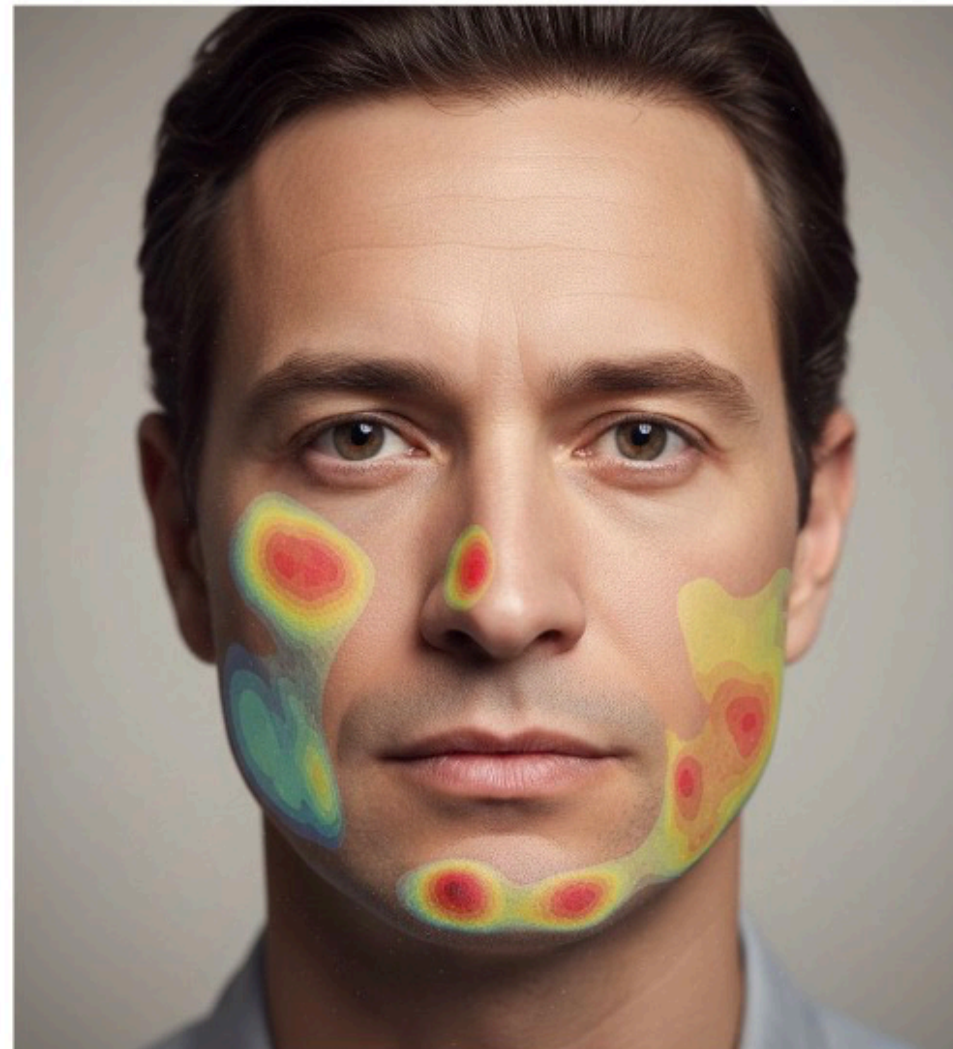


Dual Domain Analysis of Adversarial Sensitivity for Forgery Localization



Trends and Applications of Computer Vision - UniTN, 2025

Francesco Bassignana
Marco Gandolfi
Davide Vecchi

Introduction

Objectives

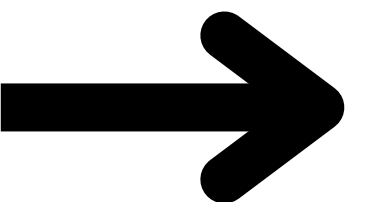
- **Standard deepfake detectors:** estimate the likelihood that an image is manipulated
- **Project objective:** investigate the possibility of performing also forgery localization



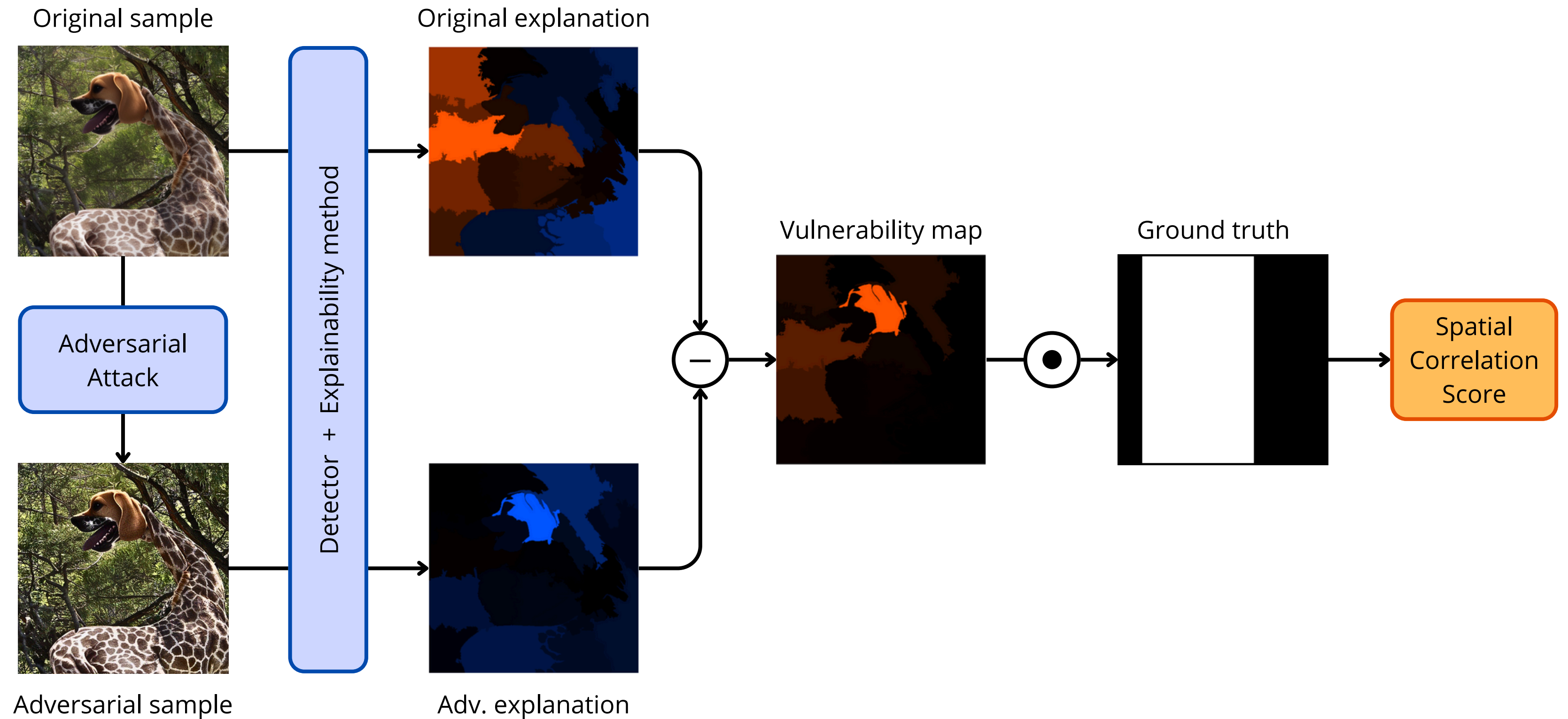
What's the Key?

What's the Key?

Adversarial Attacks!

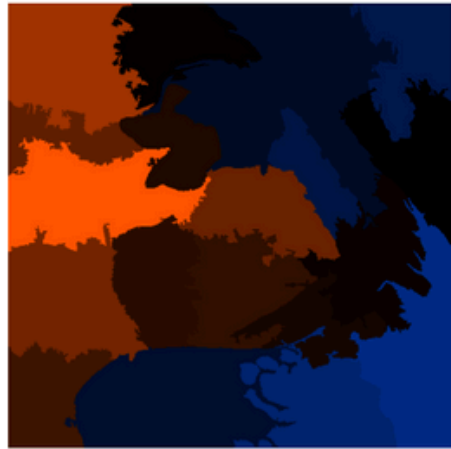


Framework pipeline



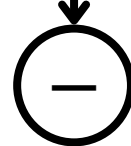
Why subtraction?

Original explanation



1) Where the model changed its activations the most within the image under adversarial attack?

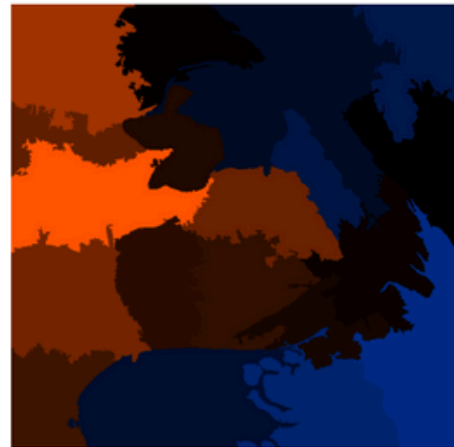
Vulnerability map



Adv. explanation

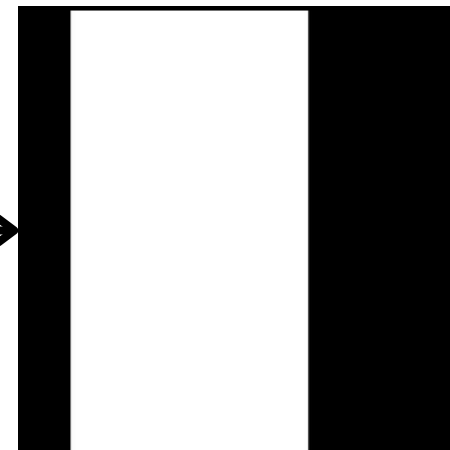
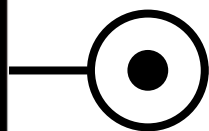
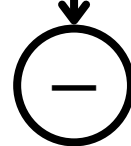
Our hypothesis

Original explanation



1) Where the model changed its activations the most within the image under adversarial attack?

Vulnerability map



Spatial
Correlation
Score



Adv. explanation

2) Are these locations coinciding with the inpainting areas?

Components Breakdown

Which images? B-Free

<https://grip-unina.github.io/B-Free/>

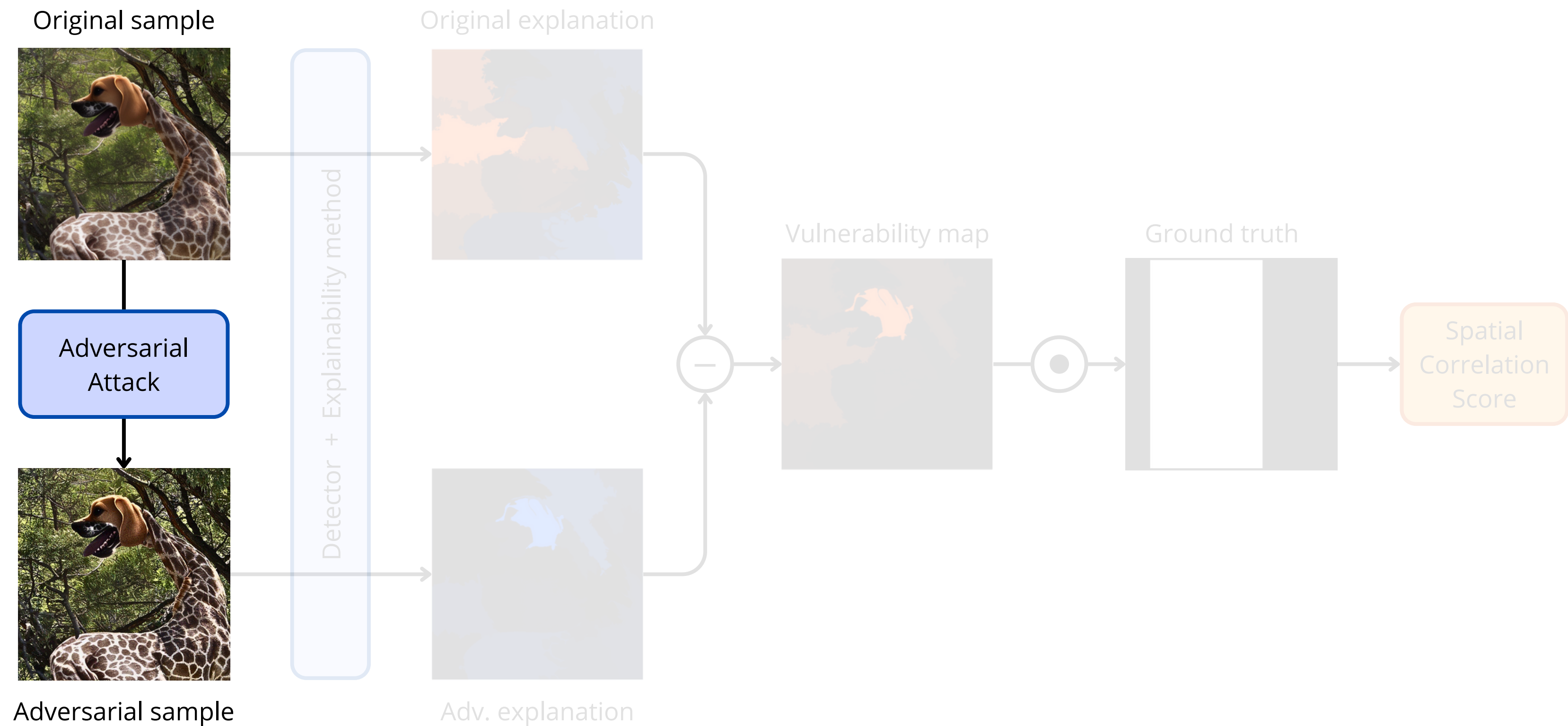
A Bias-Free Training Paradigm for More General AI-generated Image Detection

Fabrizio Guillaro¹ Giada Zingarini¹ Ben Usman² Avneesh Sud²
Davide Cozzolino¹ Luisa Verdoliva¹

¹University Federico II of Naples ²Google DeepMind



Adversarial Attacks

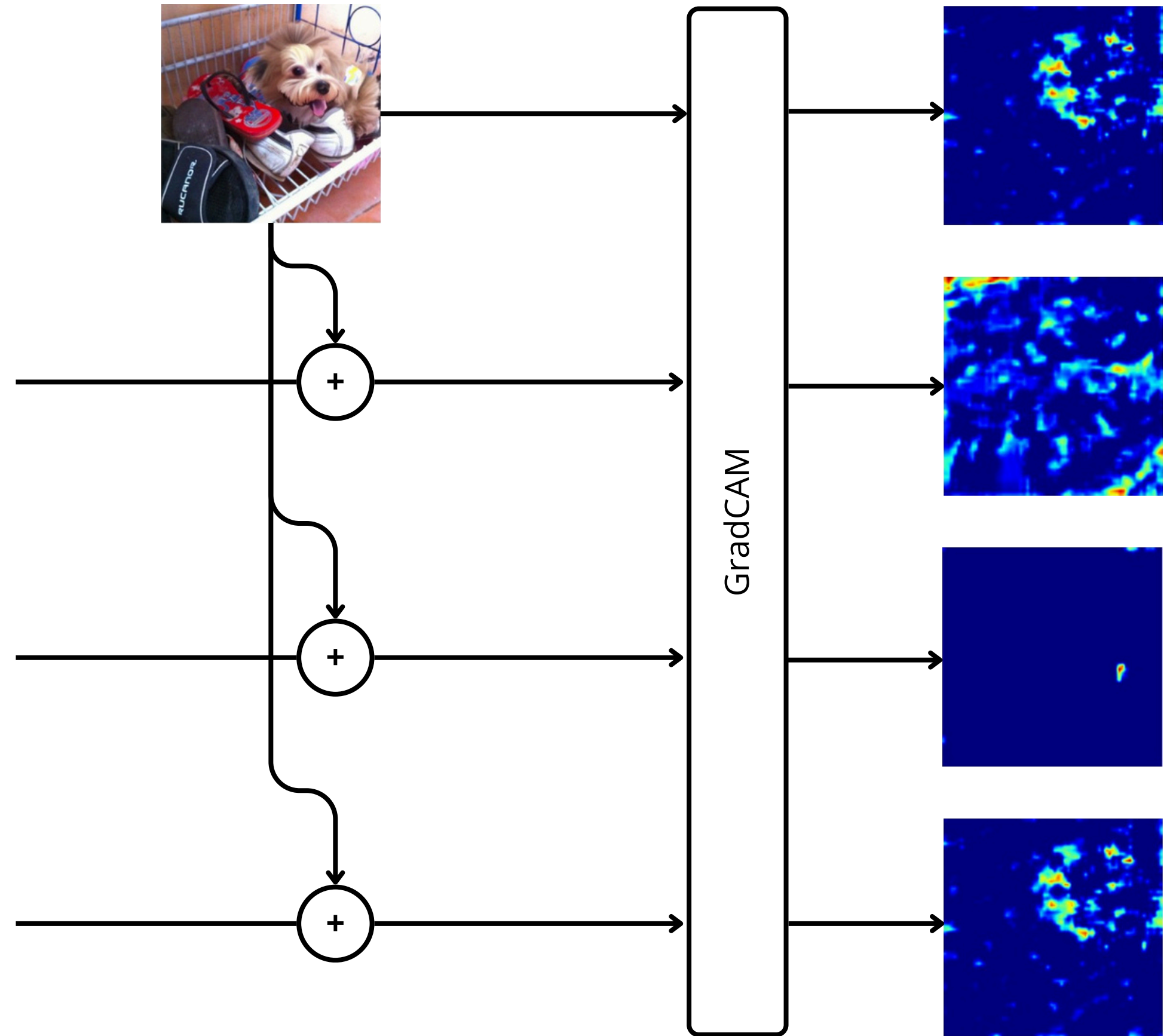


Adversarial Attacks

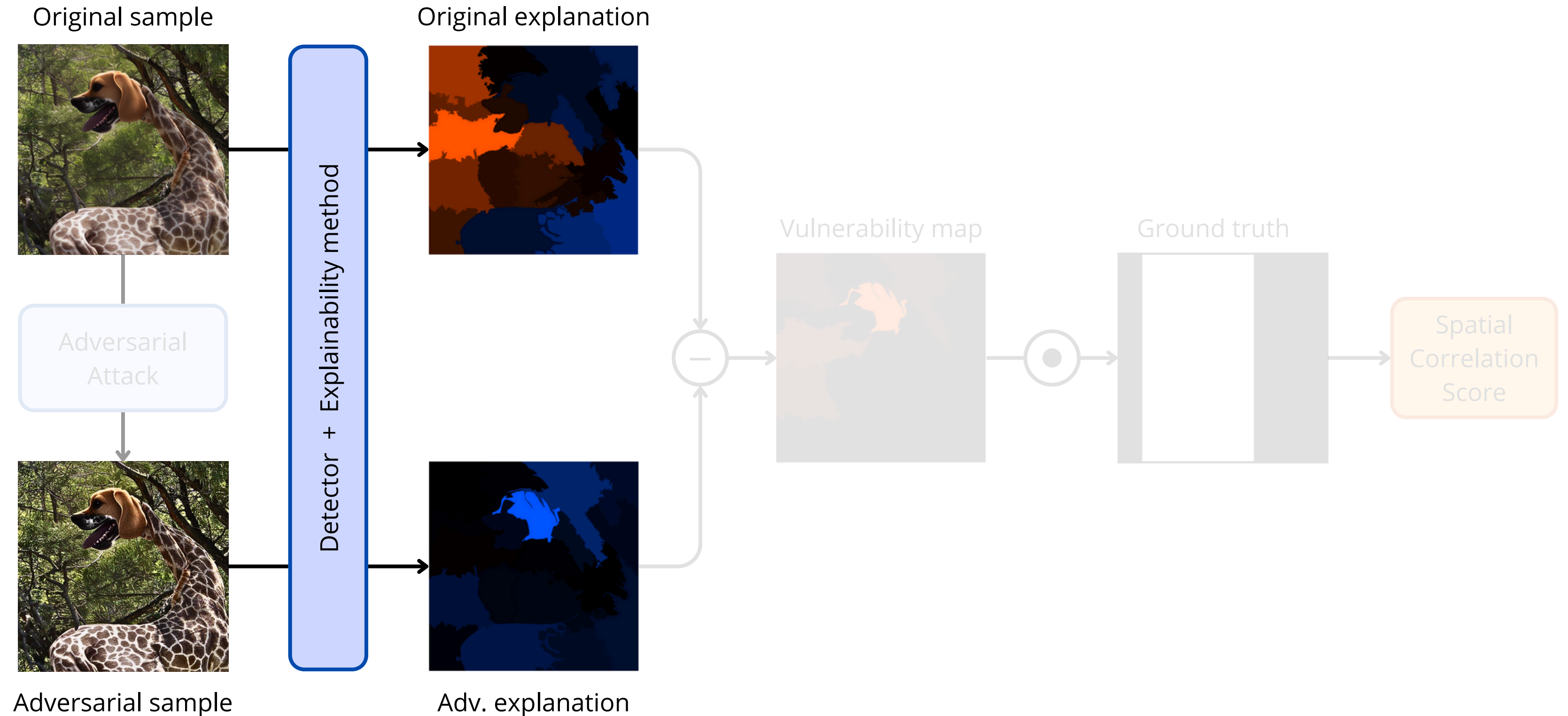
Fast Gradient Sign Method (FGSM): single-step input perturbation

Projected Gradient Descent (PGD): iterative application of FGSM

DeepFool: estimated minimum noise to flip model's prediction



Framework pipeline - Detectors and Explanations



Explainability methods

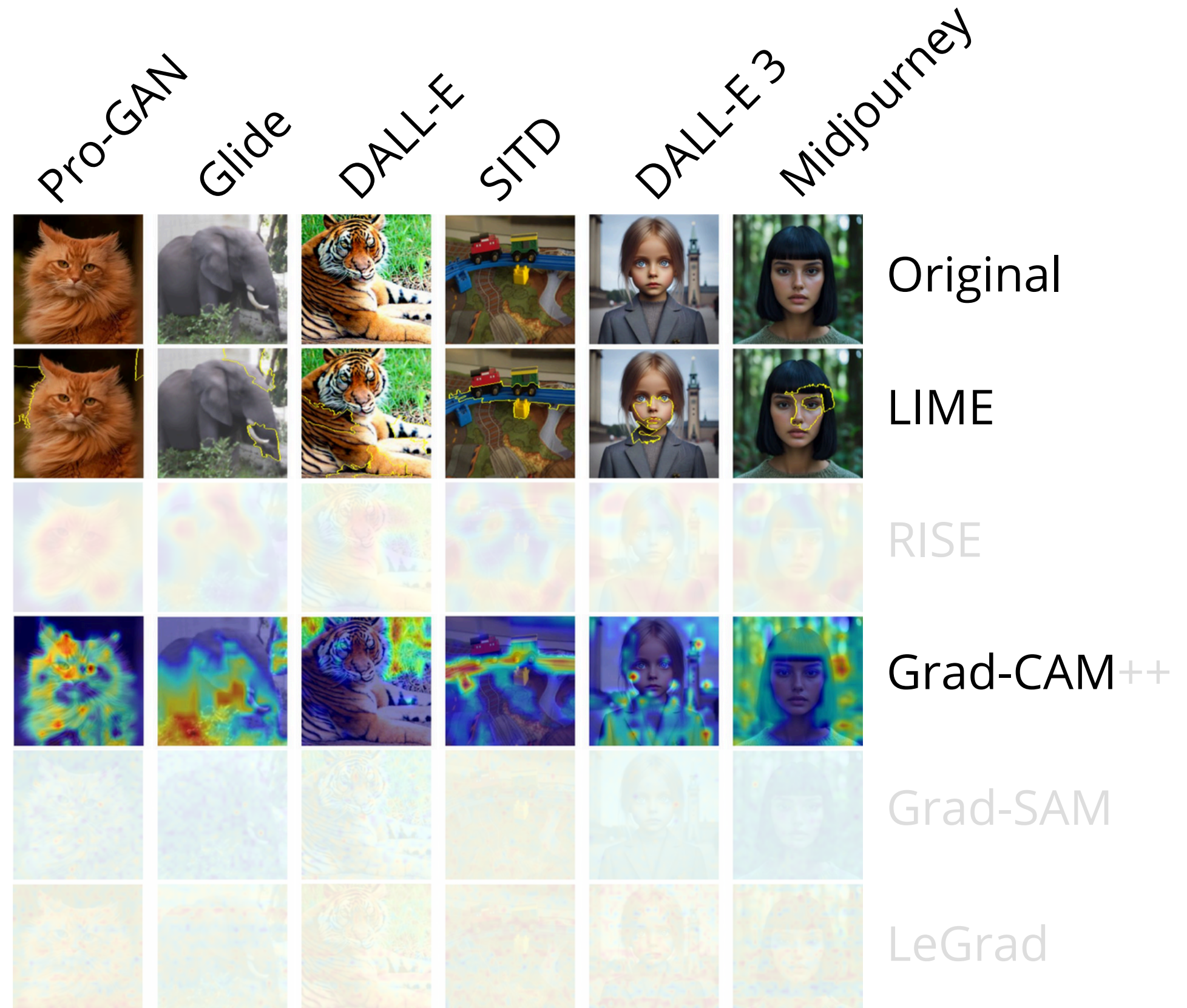
The most relevant parts of an image (highest attribution score) for the classification output act as a proxy for the vulnerability map

RESEARCH

Open Access

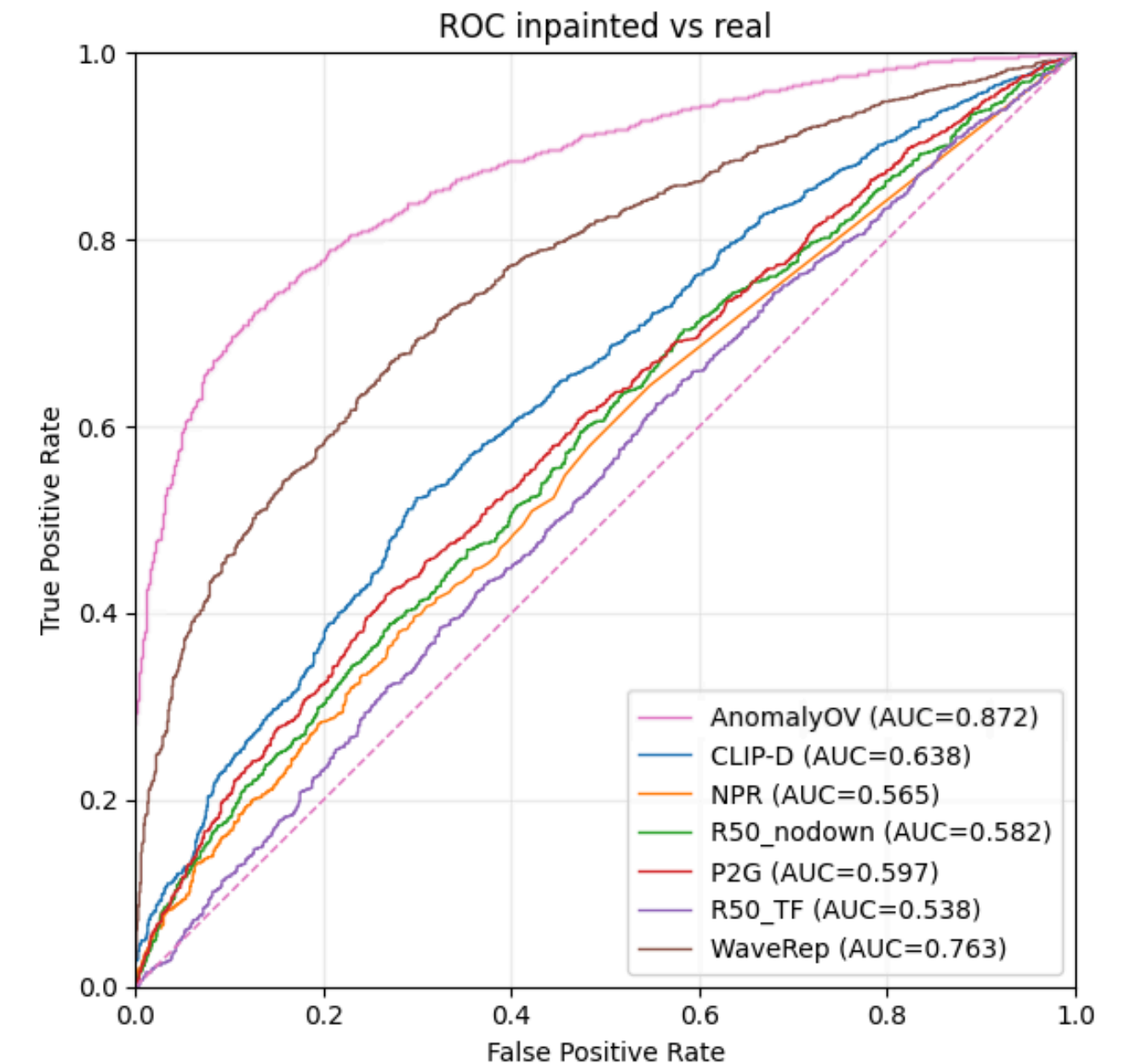
Explainability-driven adversarial robustness assessment for generalized deepfake detectors

Lorenzo Cirillo^{1†}, Andrea Gervasio^{1†} and Irene Amerini^{1†}



Pre-trained detectors

Detector	Architecture	Explanation Method
ResNet-50 (No Downsampling)	CNN (ResNet-50)	GradCAM
WaveRep	ViT (DINOv2)	LIME
AnomalyOV	Hybrid (SigLIP + anomaly expert)	Internal Saliency Map



<https://github.com/honda-research-institute/Anomaly-OneVision>

<https://github.com/grip-unina/WaveRep-SyntheticVideoDetection>

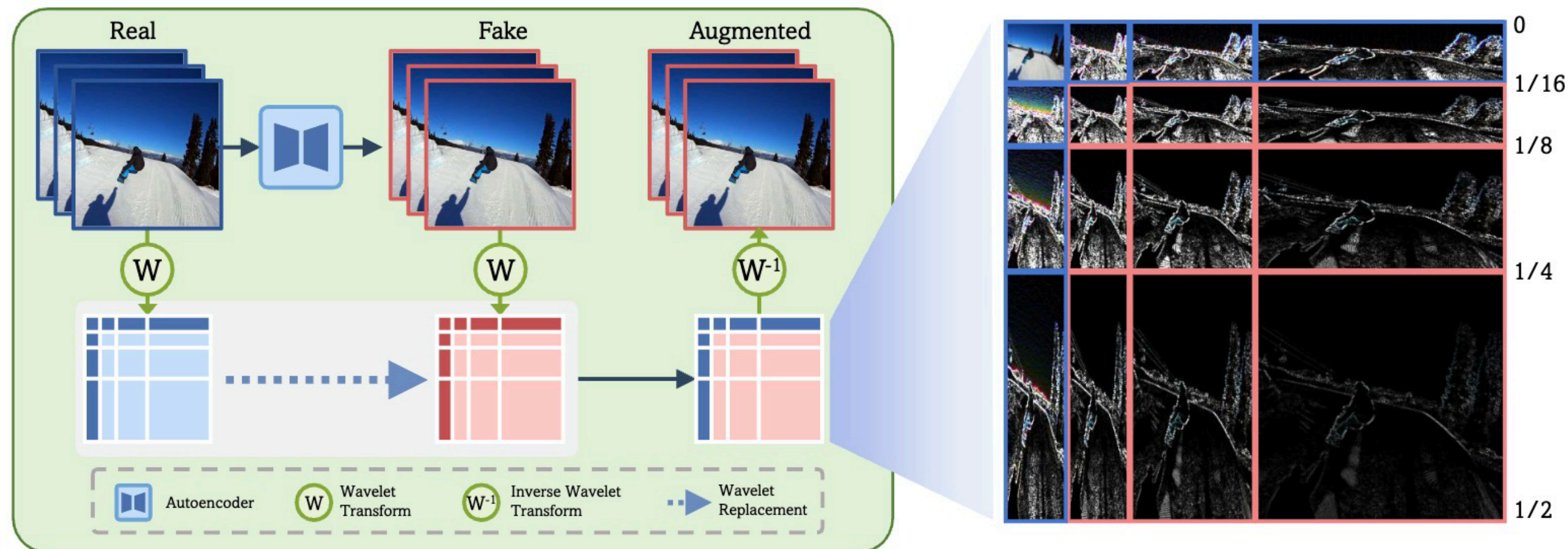
<https://github.com/truebees-ai/Image-Deepfake-Detectors-Public-Library>

WaveRep + LIME

- Pretrained DINOv2 (WaveRep, 29 Nov 2025).
- Trained with novel augmentation framework:
 - Wavelet decomposition +
 - Selective subband replacement on paired real/fake frames.

Seeing What Matters: Generalizable AI-generated Video Detection with Forensic-Oriented Augmentation

Riccardo Corvi^{1,2}, Davide Cozzolino¹, Ekta Prashnani²,
Shalini De Mello², Koki Nagano², Luisa Verdoliva¹
¹University Federico II of Naples, ²NVIDIA

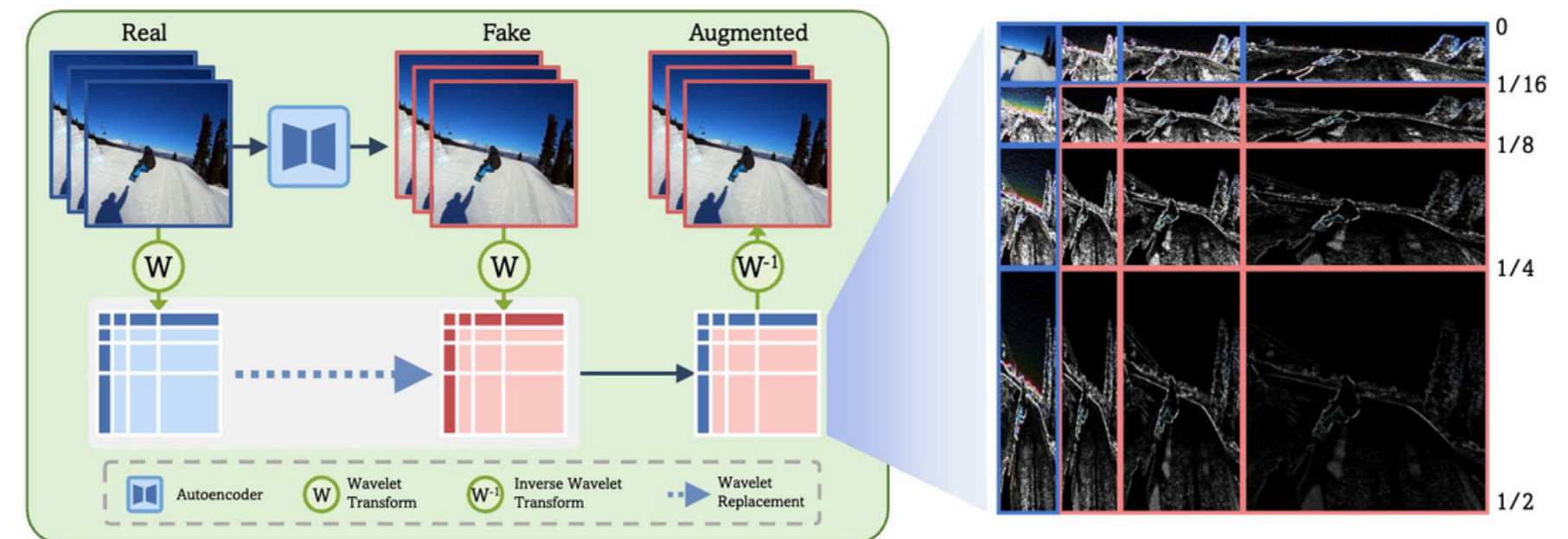


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- Pretrained DINOv2 (WaveRep, 29 Nov 2025).
- Trained with novel augmentation framework:
 - Wavelet decomposition +
 - Selective subband replacement on paired real/fake frames.
- Benefits: decision rely on robust diagonal mid-high frequency forensic cues.
 - Better cross-generator generalization.
 - Robust to compression artifacts.
- Drawbacks:
 - Wavelet cues are sparse →
 - Gradient based explanations fail.
- Solution: use perturbation-based explanations.



Anomaly OV in a nutshell

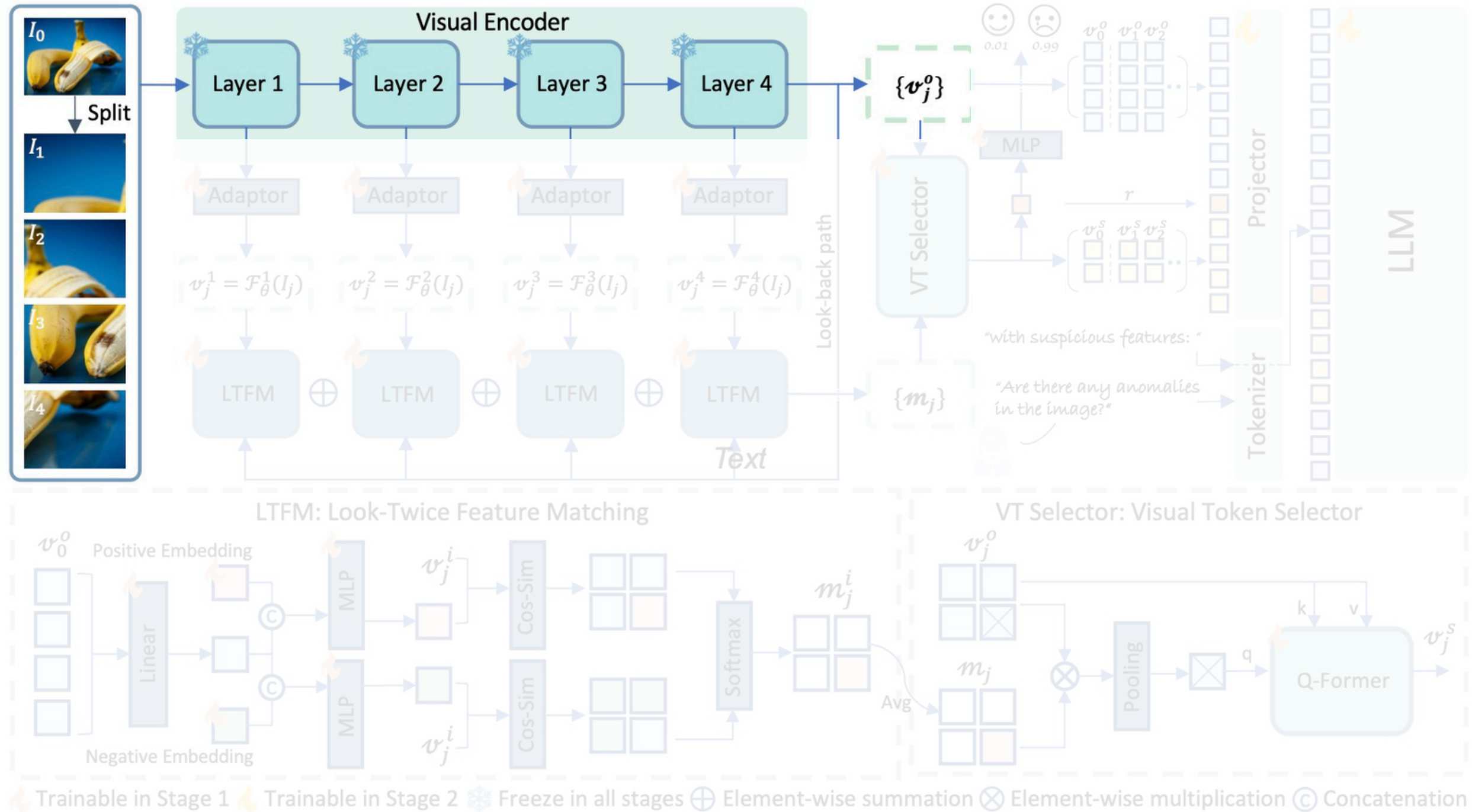


Figure 3. Overview of the **Anomaly-OV** architecture. It consists of two training stages: (1) professional training for the anomaly expert, and (2) visual instruction tuning for anomaly detection and reasoning. Text and visual tokens are distinguished by different colors.

Anomaly OV in a nutshell

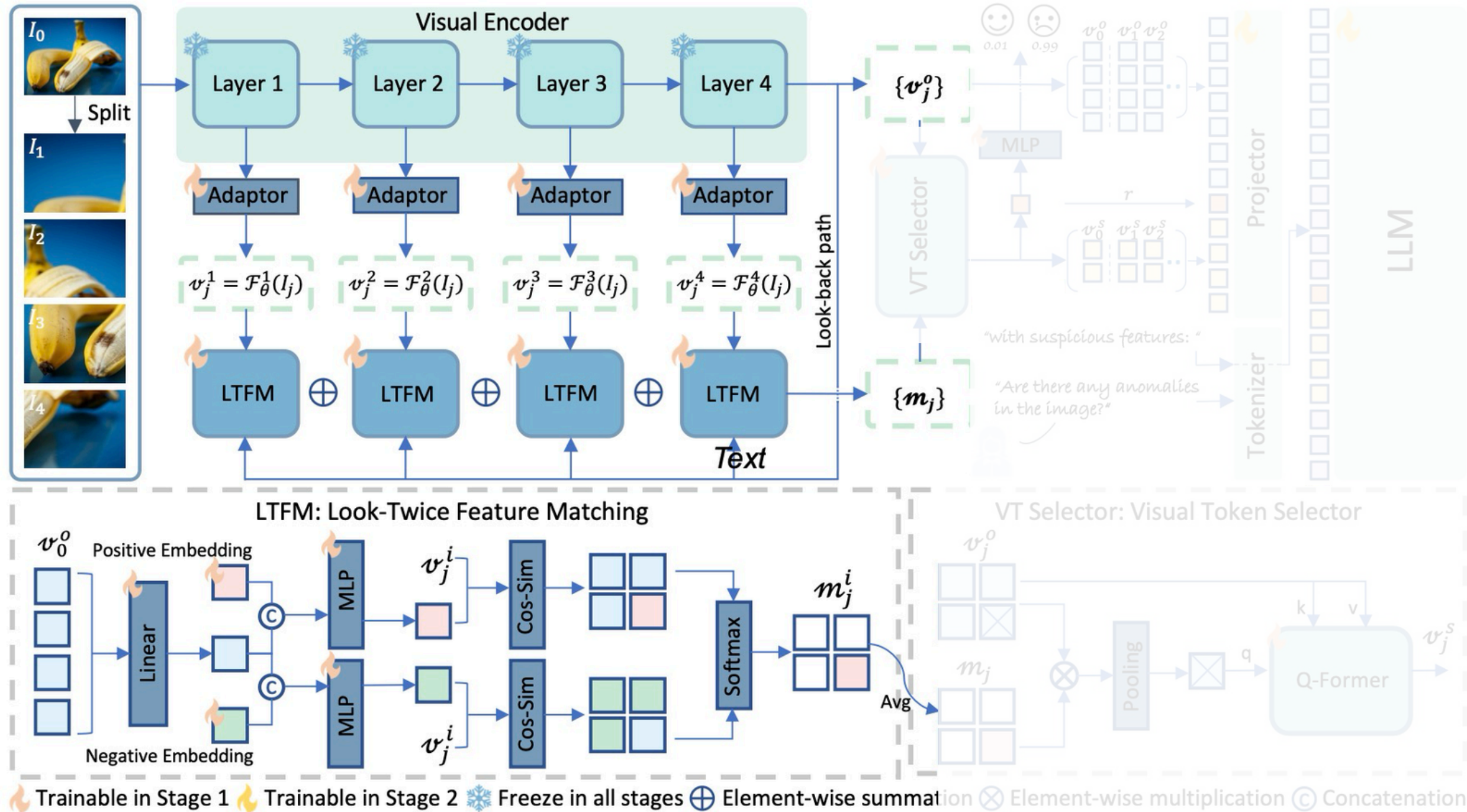


Figure 3. Overview of the **Anomaly-OV** architecture. It consists of two training stages: (1) professional training for the anomaly expert, and (2) visual instruction tuning for anomaly detection and reasoning. Text and visual tokens are distinguished by different colors.

Anomaly OV in a nutshell

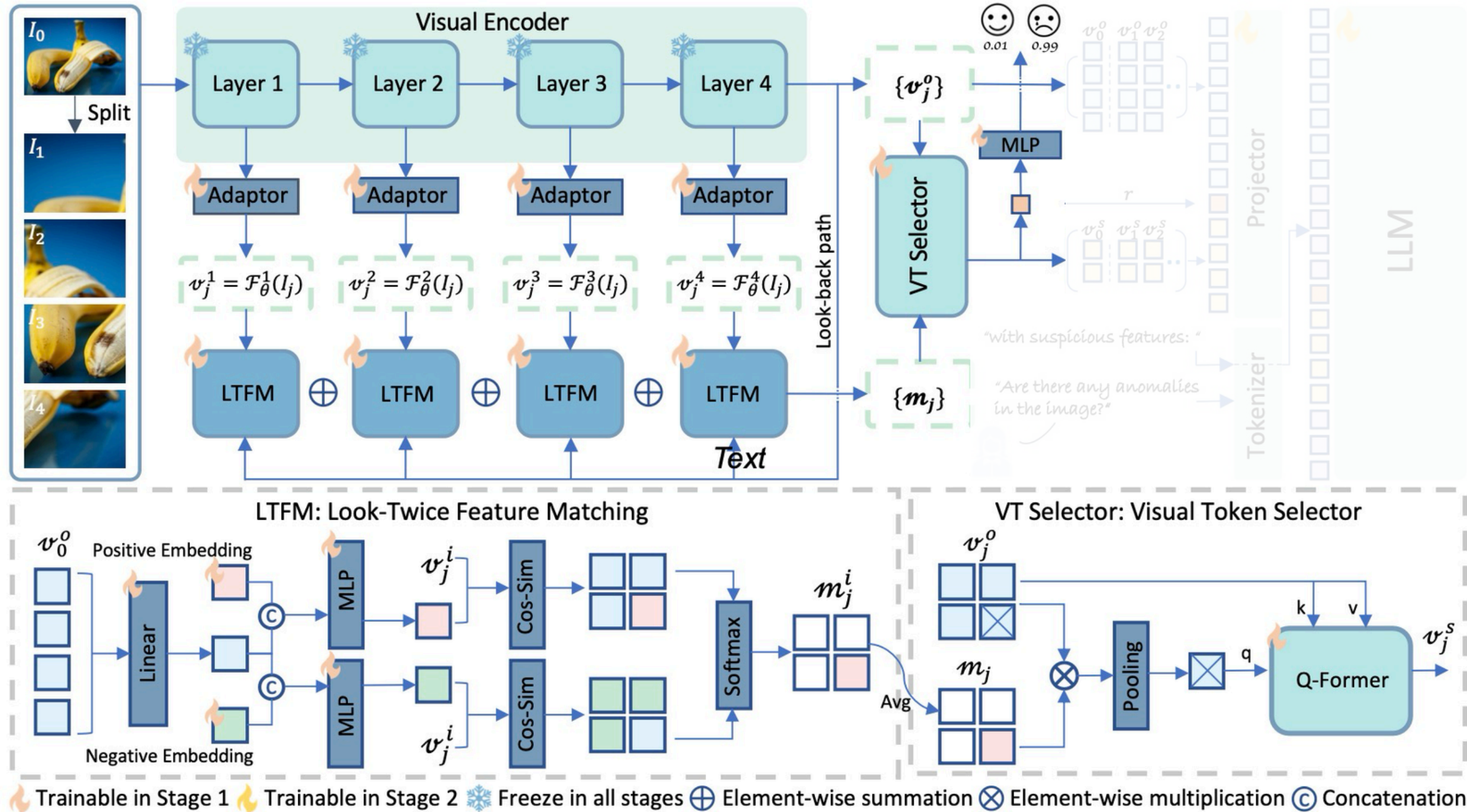
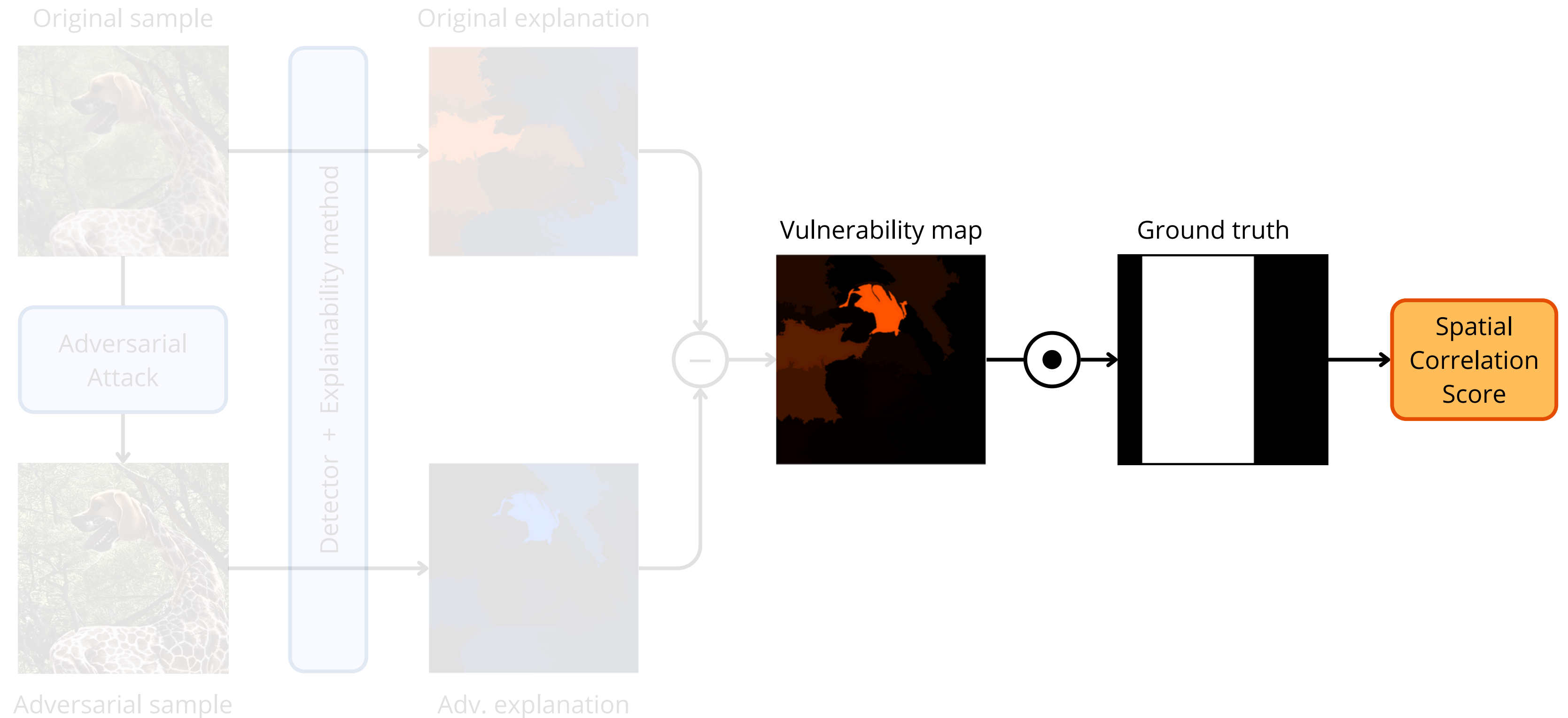


Figure 3. Overview of the **Anomaly-OV** architecture. It consists of two training stages: (1) professional training for the anomaly expert, and (2) visual instruction tuning for anomaly detection and reasoning. Text and visual tokens are distinguished by different colors.

Framework pipeline - Vulnerability Analysis

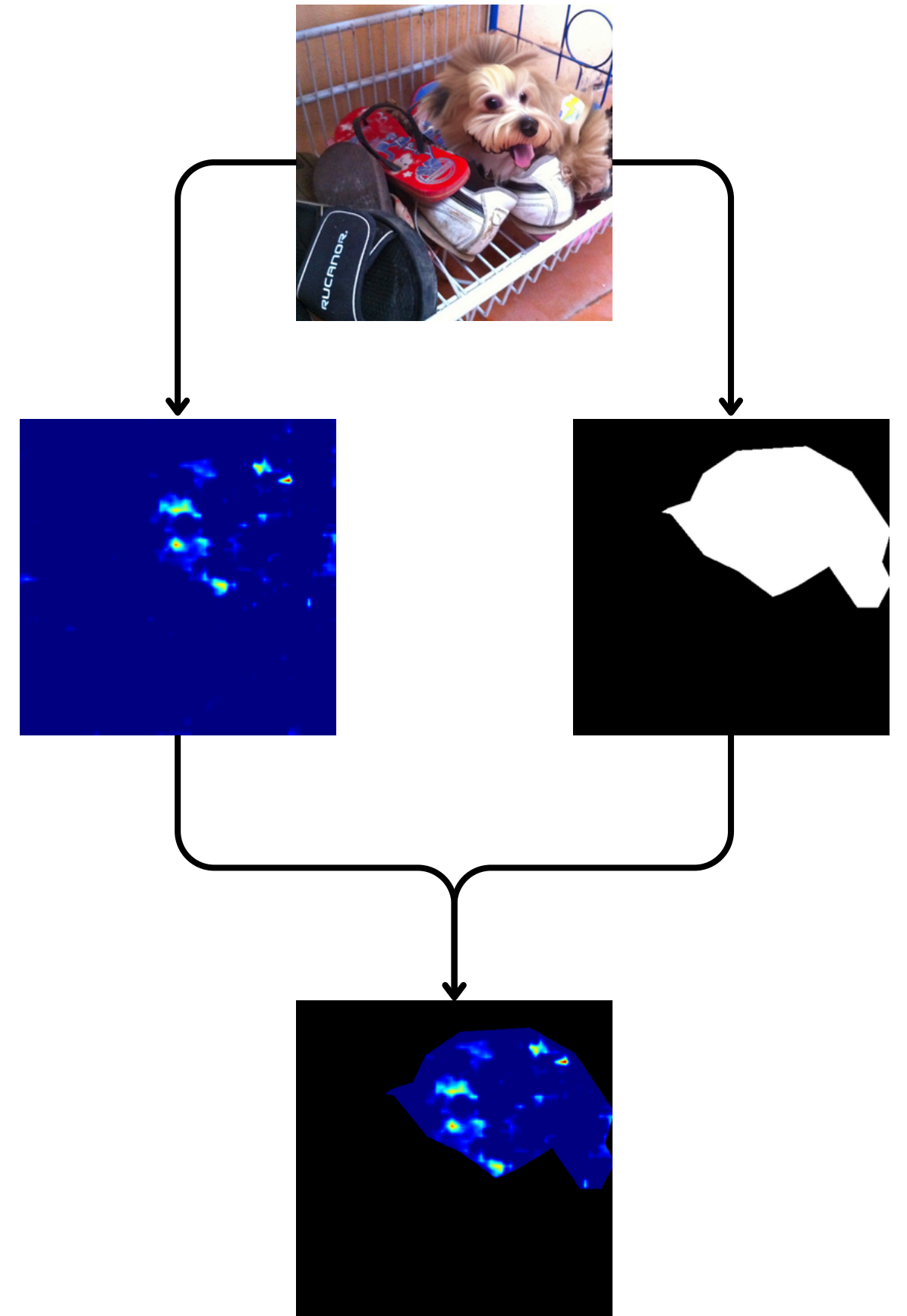


Final evaluation metrics

Pixel-wise classification

$$\text{ROC-AUC} = \text{AUC}(\text{TPR}(\tau), \text{FPR}(\tau))$$

$$\text{PR-AUC} = \text{AUC}(\text{Precision}(\tau), \text{Recall}(\tau))$$



Final evaluation metrics

Pixel-wise classification

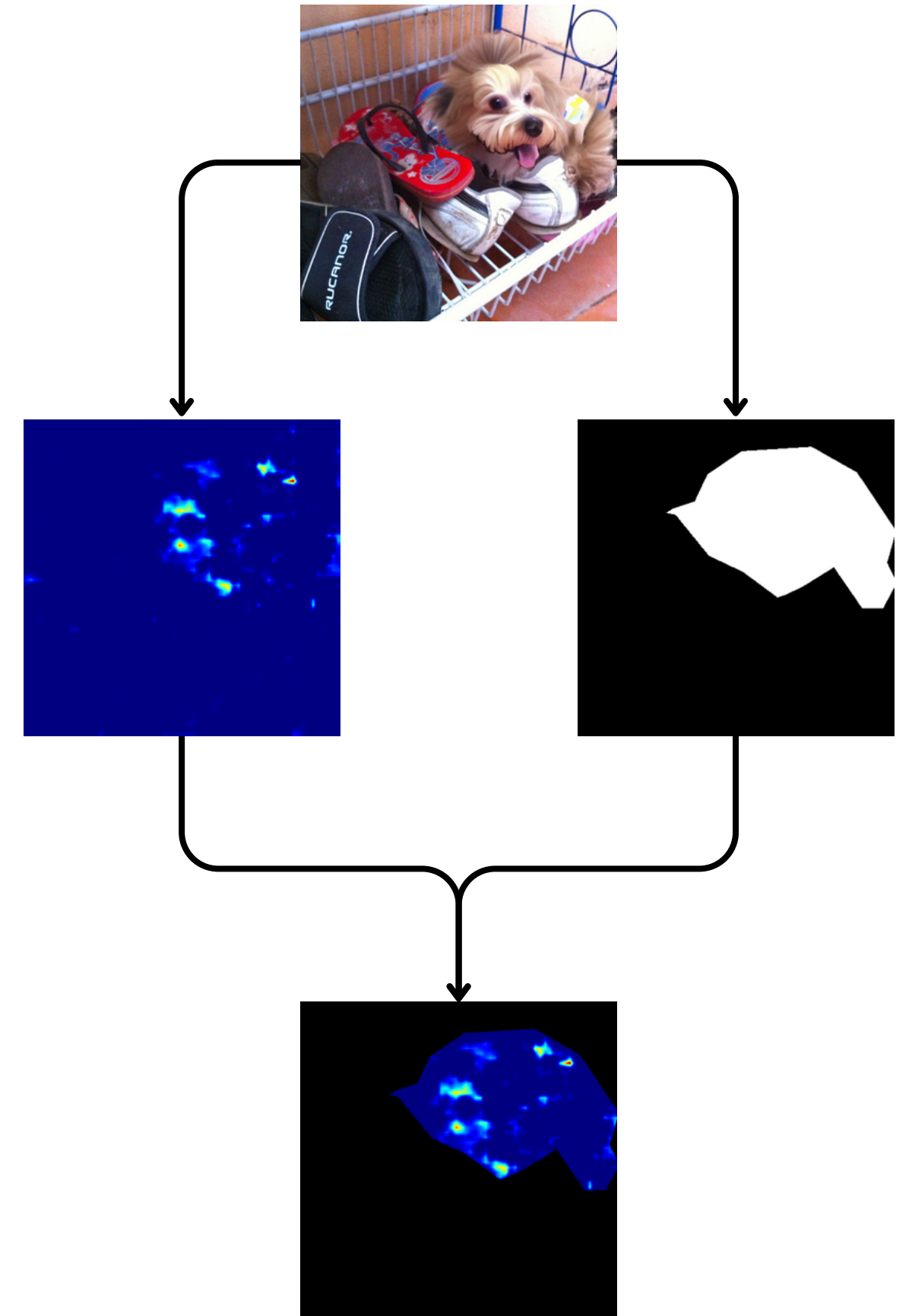
$$\text{ROC-AUC} = \text{AUC}(\text{TPR}(\tau), \text{FPR}(\tau))$$

$$\text{PR-AUC} = \text{AUC}(\text{Precision}(\tau), \text{Recall}(\tau))$$

Spatial Correlation

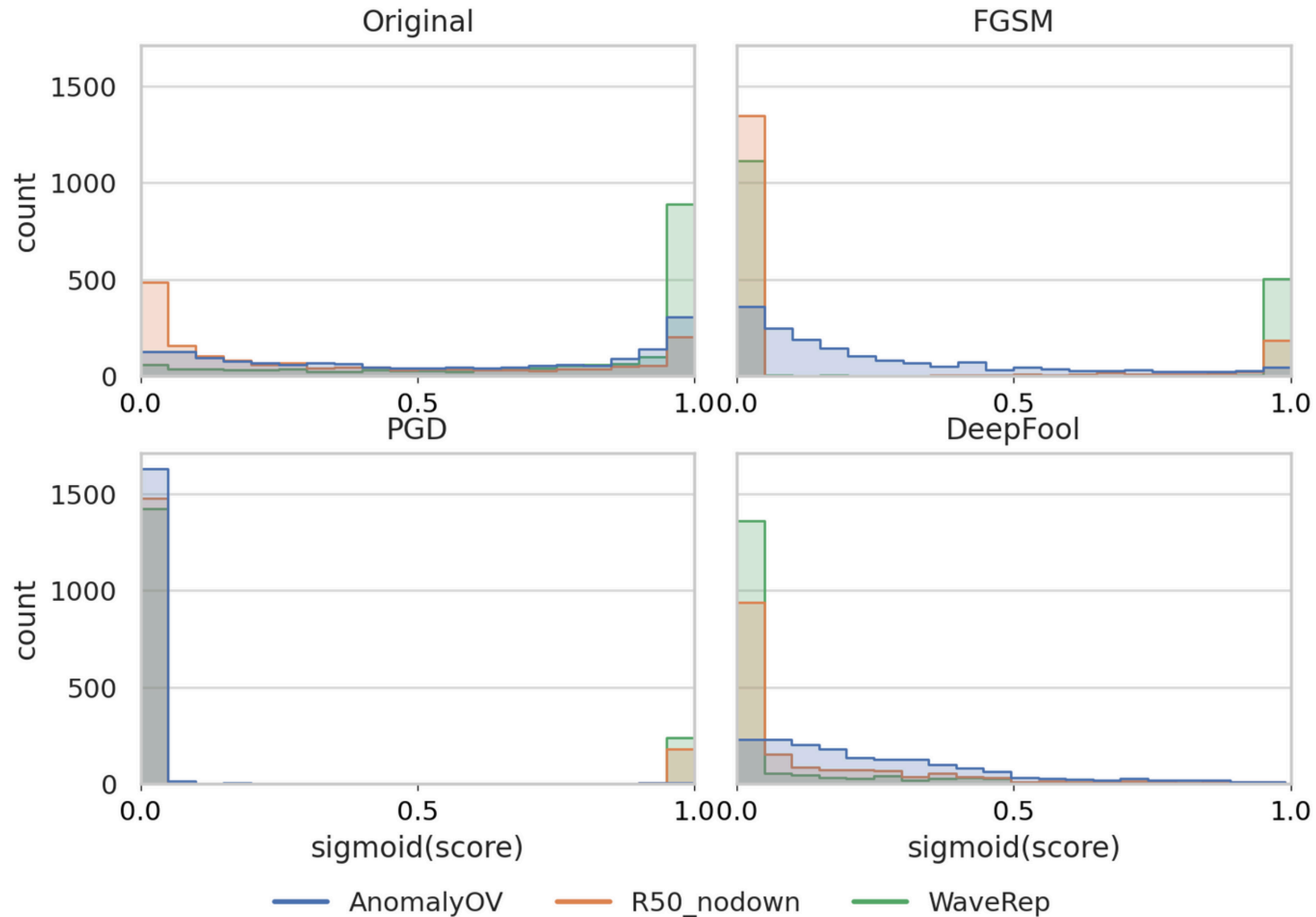
$$\text{MF}(S, M) = \frac{\sum_{i \in M} S_i}{\sum_i S_i}$$

$$\text{IoU@Top-}k = \frac{|\text{Top-}k(S) \cap M|}{|\text{Top-}k(S) \cup M|}$$

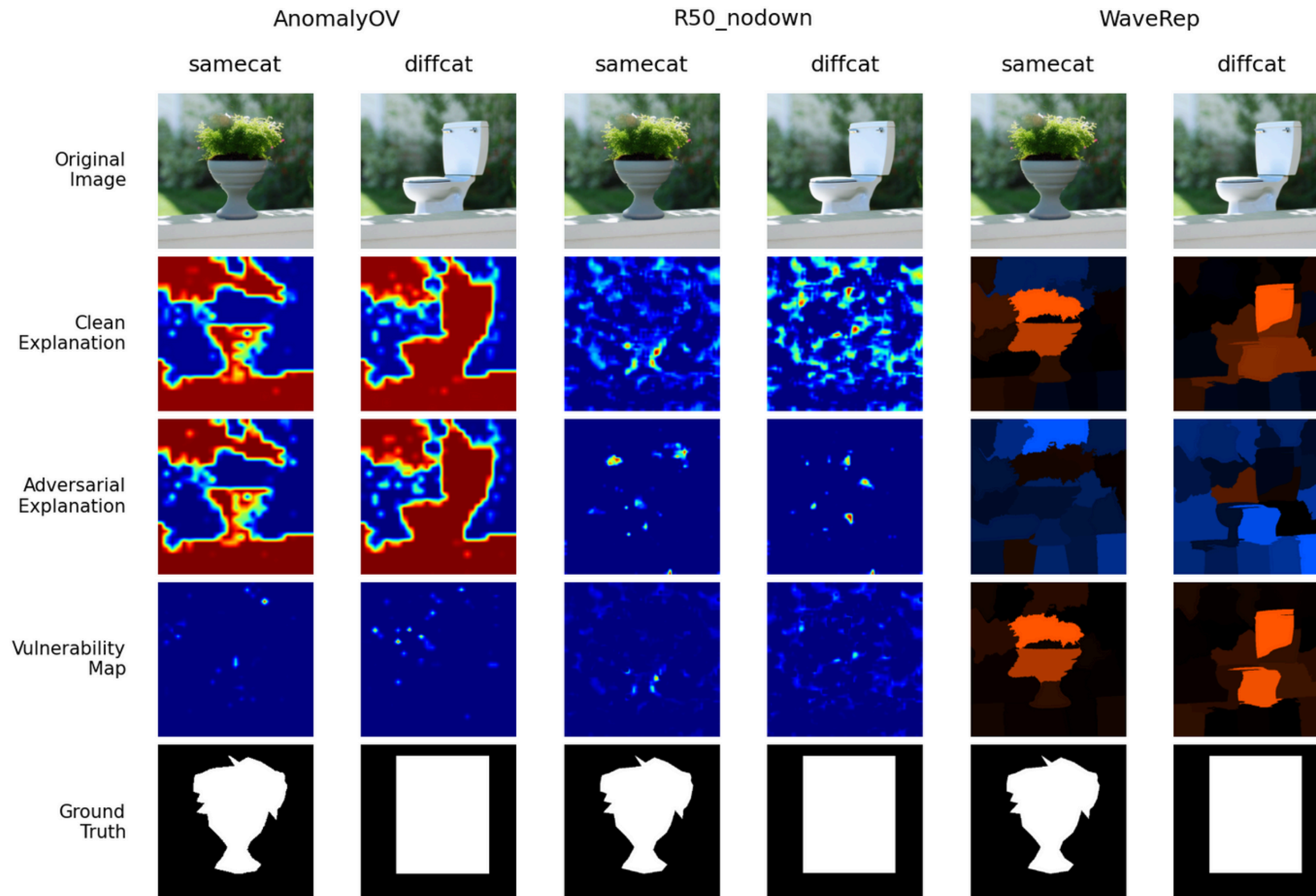


Results

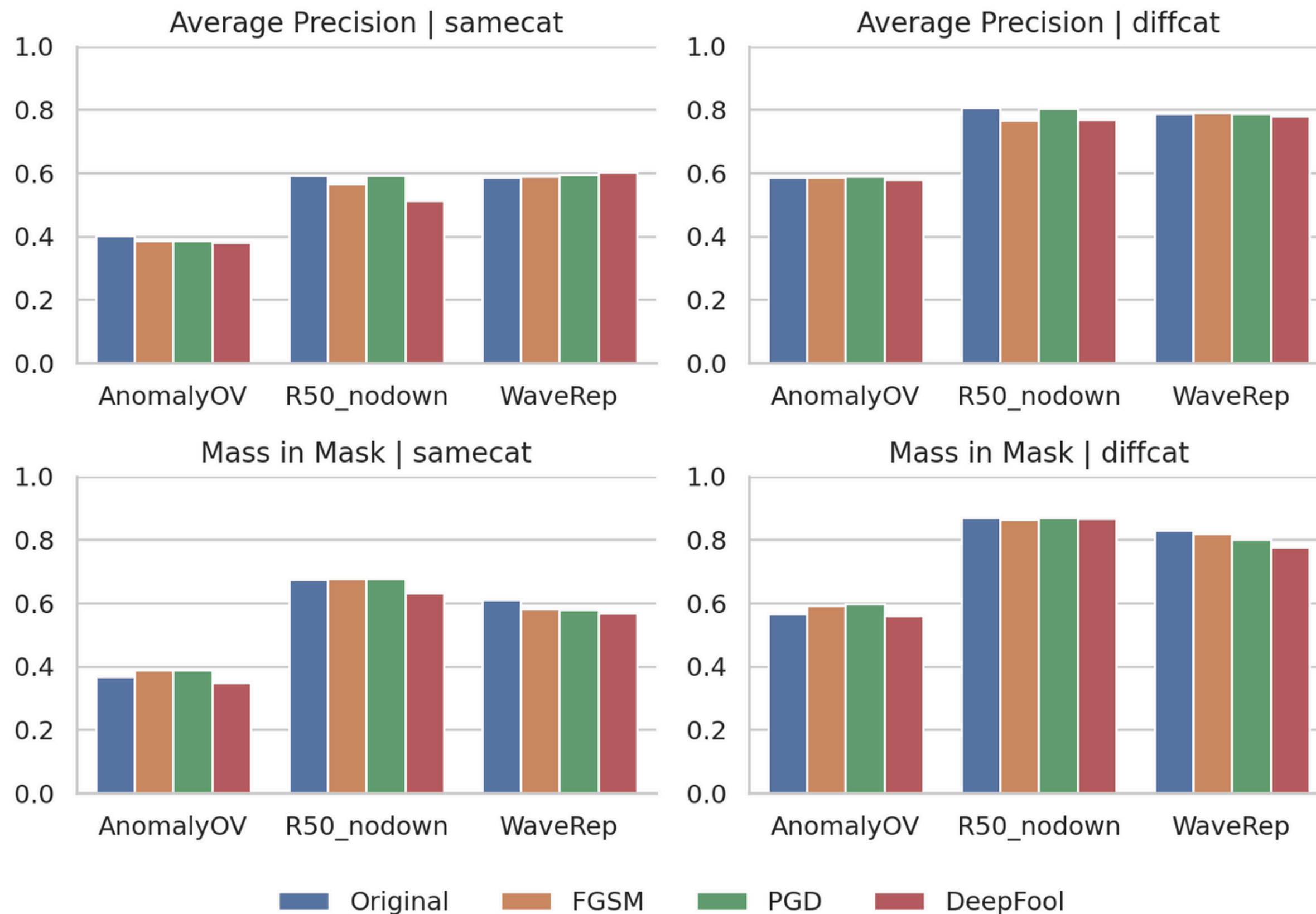
Effects of Attacks



Qualitative Comparison

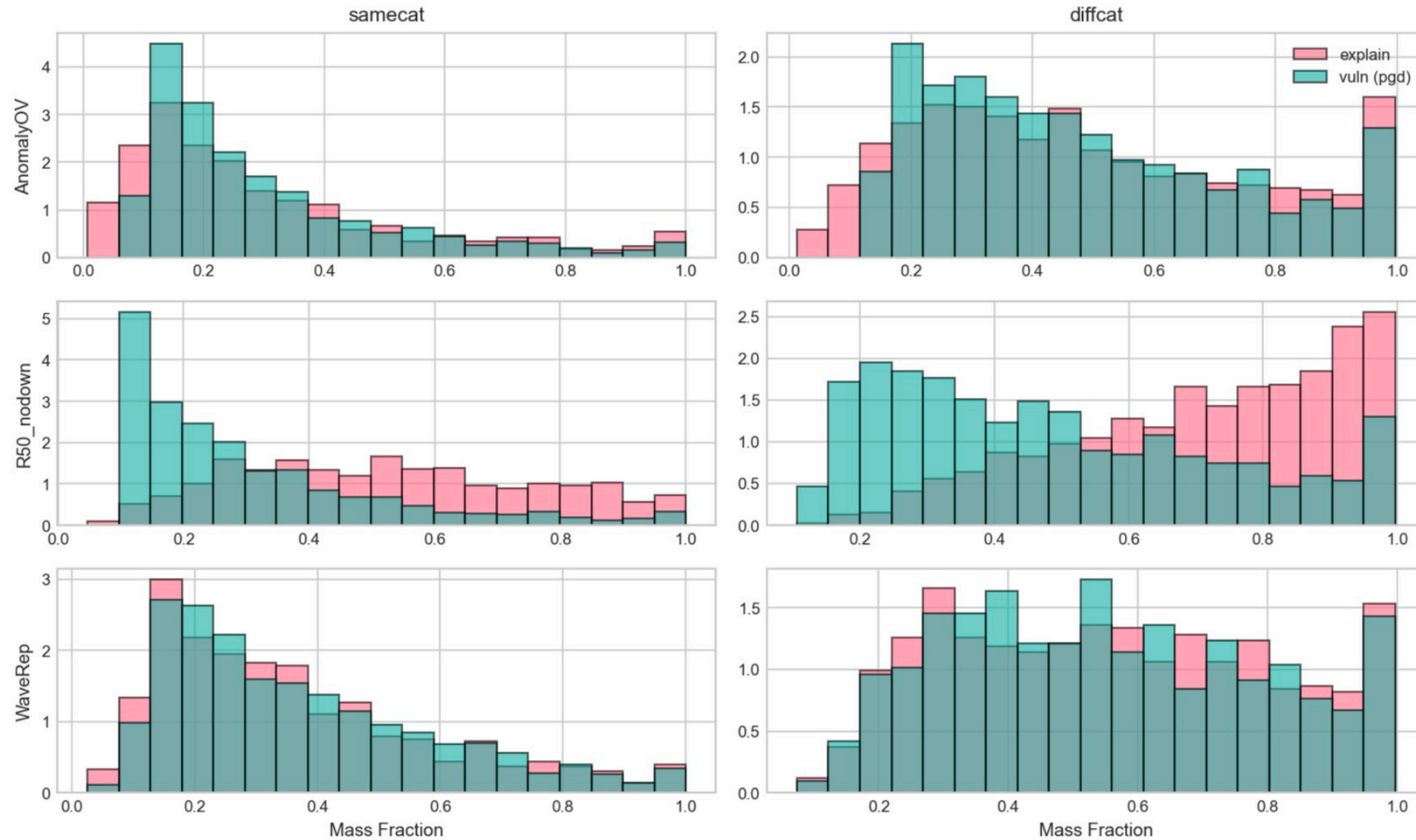


Quantitative Comparison



Mass Fraction Distribution

Explanation vs Vulnerability (Mass Fraction)



Conclusions

- Adversarial vulnerability maps yield reliable forgery localization.
- Consistent with clean explainability, robust across attacks.
- Architecture matters most, best with spatially biased detectors.
- Amplify existing spatial sensitivity (not new cues).
- Complementary tool to standard explainability.

Future Directions

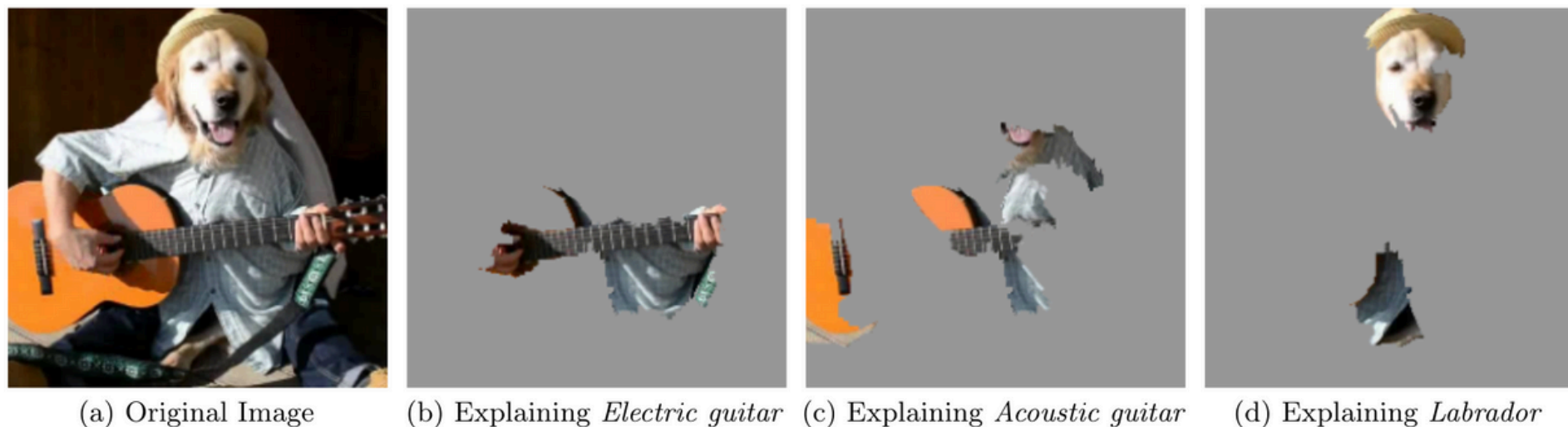
Vulnerability Maps 3 - Random Perturbations

Idea:

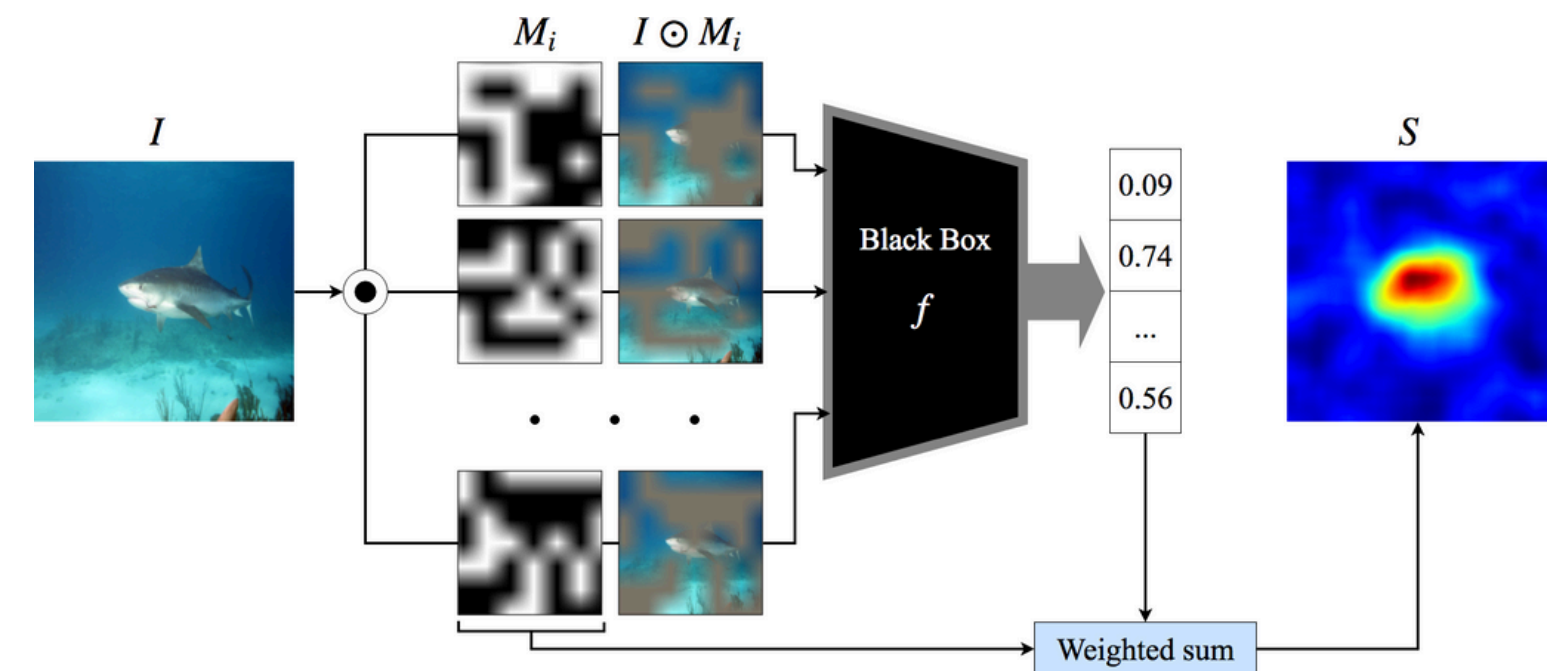
- Obtain the vulnerability maps by randomly attacking parts of the images.
- Use adversarial noise instead of gray patches in perturbation based explainability models (e.g. LIME and RISE).

Pro: effectively explains the vulnerable areas under random local attacks.

Con: high computational cost (one attack per iteration).



Local Interpretable Model-agnostic
Explanations (LIME) **[4]**



Randomized Input Sampling for
Explanation (RISE) **[5]**

Limitations and Future Works

Known limitations	Future improvements
Limited set of detectors/explainability methods	Extend the existing model library
AnomalyOV's bias towards anomaly detection/semantic saliency	Train anomaly expert from scratch
Frequency-domain contribution not considered	Investigate the influence of manipulations in the frequency domain

Thanks for the attention!

Questions?